Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/08/2024
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Spaceflight Effects on Plant-Microbe Interactions		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	04/01/2021	**End Date:**	03/31/2025
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Romeyn, Matthew	**Contact Phone:**	321-867-6025
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 03/31/2025 per NSSC information (Ed., 4/8/24).		
Key Personnel Changes/Previous PI:	NA		
COI Name (Institution):	Bakshi, Arkadipta Ph.D. (University of Wisconsin, Madison) Swanson, Sarah Ph.D. (University of Wisconsin, Madison) Barker, Richard Ph.D. (University of Wisconsin System) Hanson, David Ph.D. (University of New Mexico)		
Grant/Contract No.:	80NSSC21K0577		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This work seeks to address: (1) how spaceflight modulates the interactions between plants and microbes and (2) how well microgravity analogs capture the events elicited by the spaceflight environment. Tomato plants will be grown on orbit in the NASA Vegetable Production System (Veggie) hardware on board the International Space Station (ISS) with and without the beneficial rhizosphere microbe Trichoderma hazianum. A third sample will be of this microbe growing under identical conditions on the ISS but without the plants. These samples will be compared to parallel ground controls at 1 x gravity as well as to samples growing on 1-axis and 3D clinostats. Assays will integrate RNAseq-based transcripomics and ionomics (nutrient uptake and distribution) alongside biochemical measures of photosynthesis and stress. These comparisons will provide measures of both how spaceflight affects the plant, the microbe, and the relationship between these organisms and additionally, how well microgravity analogs can reproduce these kinds of events on the ground. In addition, the omics-level data gathered from this study will be compared to the wealth of spaceflight-related omics data available through the GeneLab data repository. Using an approach of orthologous matrix mapping will allow identification of similar genes between diverse species and so allow for comparisons of, for example, the degree of similarity between patterns of gene expression to be compared between different species. Overall this research will help define how spaceflight may modulate plant, microbial, and plant-microbe responses and help understand whether defined beneficial microbes may provide a countermeasure to the deleterious effects of spaceflight on plants. The work will capitalize on the complementary expertise of two groups: the Gilroy lab team (plant spaceflight, transcriptomics) and the Hanson lab (biochemistry, photosynthesis).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposed research seeks to address how spaceflight modulates the interactions between plants and microbes using tomatoes and the beneficial soil microbe Trichoderma harzianum. The microbiome around the plant root is recognized as a crucial element in the productivity and hardiness of plants but the complex interactions and chemical signals that occur between plant and microbe have only recently begun to be dissected. Yet, for example, T. harzianum is used as a commercial biostimulant, being added to the soil to pormote plant growth and vigor. The research in this project will further define the molecular components of the interaction between plant root and this fungus and how spaceflight alters these events. Thus, the work will not only provide insight into how plant-microbe interactions are affected by spaceflight but address whether T. harzianum might be used as a biostimulant to counteract some of the stresses of spaceflight on plant growth. This research will also help provide molecular insight into how these interactions occur on Earth. Such a fuller understanding of plant-microbe interactions, especially of T. harzianum-plant root communications and response will be important steps towards optimizing these beneficial interactions and so increasing plant productivity in both space and on Earth.
Task Progress:	The major focus for this work to date has been to define experiment protocols and analyses through a Science Verification Test (SVT) and Experiment Verification Test (EVT). These rounds of testing and refinement have then allowed conducting the flight experiment. The SVT was completed in June of 2022 and EVT in October 2022. A further testing of how to perform the harvesting protocols in the Life Science Glovebox was conducted at NASA Johnson Space Center in March 2023. The glovebox was required to ensure that the fungal samples were contained away from the crew and cabin environment during crew harvest procedures as the Petri dishes that normally contained these samples had to be opened to facilitate harvesting. All of these tests were successful with a rating of excellent across all success criteria, allowing for the experiment to proceed to flight at the beginning of 2024. The experiment was successfully launched on January 30th, 2024 on the NG-20 Cygnus spacecraft. The biological samples consisted of sterile Petri plates containing nutrient gel medium with either tomato seeds alone, tomato seeds inoculated with spores of the beneficial, symbiotic fungus Trichoderma harzianum, or Petri plates inoculated with Trichoderma alone. A 2 mm pore size polyester mesh that was bounded by a thin 3 mm wide, 0.5 mm thick, 3d printed PETG perimeter rim was incorporated onto the top of the nutrient gel before planting of the seeds or spores to facilitate crew harvesting of plants and fungal colonies at the end of the experiment. These experiment samples were prepared at NASA Kennedy Space Center and then transported to the International Space Station (ISS) wrapped in foil. Transport was at 4°C to prevent premature seed or fungal spore germination. Upon reaching the ISS on February 5th 2024, the samples were unwrapped and installed in the growth chamber hardware consisting of two Veggies. [Ed. Note: Veggie" is the NASA Vegetable Production System.] Plants were grown under continuous light for 14 days, with photography by the crew every other day. These images showed that there was no seed germination or visible fungal growth from the Trichoderma at the point of installation in the Veggie. In contrast, two days later, the warming to cabin temperature and exposure to light led to seed germination, as evidenced by obvious visible protrusion of the embryonic root from the seeds. A small ~1 cm diameter hyphal mat was also evident radiating from either the site of inoculation on the Trichoderma only plates or from the seeds in the plates where the tomato seeds had been inoculated with Trichoderma spores. As the experiment progressed, the tomato seedlings exhibited continued root and shoot extension and developed to the stage of two expanded true leaves. An expansion of the fungal mycelium was evident, reaching the furthest extent of the Petri plate over approximately 8 days. A parallel ground control was performed in the ISS environmental simulator chambers at Kennedy Space Center. Preliminary comparison of the images of the flight samples to these ground controls suggests both flight and ground experiments had similar time courses of plant germination and fungal growth. The root system of the flight samples showed a more random directionality than the downward gravitropic responses seen in the ground controls. The flight samples also appear to show increased root branching. The experiment also included gas sampling of the air around the Veggies to allow CoInvestigator (CoI) David Hanson at the University of New Mexico to perform stable carbon isotope analysis. By combining data from stable carbon isotope analysis from the CO2 of the cabin air with isotope analysis from the plants grown in orbit, a calculation of parameters such as plant water use efficiency can be made. The cabin air was sampled at the installation in the Veggie and at the end of the experiment using Silonite coated gas sampling canisters. In addition, each Veggie had two Petri plates containing Zeolite installed. The Zeolite passively absorbs the cabin CO2, providing a second sampling strategy. At the end of the experiment, the Zeolite plates were placed in gas tight bags for sample return. Both the sampling canisters and the Zeolite plates performed as expected and were successfully stored at the end of the experiment to await sample return and ground-based isotope analysis. On day 14 (the harvest day for the plants), the Petri plates containing the plants and fungi were transferred to the Life Science Glove Box to ensure containment of any fungal materials. The biological samples (plants and/or fungus) was harvested by peeling the polyester mesh from the gel surface. The plants and fungal hyphae remained adhered to the

	mesh, facilitating sample retrieval from the nutrient gel. Each mesh sample was then wrapped in foil and rapidly frozen by clamping between aluminum blocks conditioned to -160°C. All samples were successfully harvested, frozen, and were stored at -80°C in the Minus Eighty Life Science Freezer on the ISS (MELFI) prior to sample return. All the biology samples were returned frozen, along with the gas samples, on the CREW 7 vehicle that splashed down on March 12th, 2024. All were successfully transferred to Kennedy Space Center and the gas samplers and Zeolite samples were shipped directly to the University of New Mexico for isotopic analysis. The plant and fungal samples were stored at -80°C at Kennedy Space Center prior to shipment to the University of Wisconsin-Madison for post-flight analyses. Ongoing post-flight analyses include extraction of nucleic acid to perform RNA sequencing (RNAseq) to catalog patterns of gene expression. Additionally, measurements of the levels of chlorophyll and the stress pigment anthocyanin are being performed alongside quantification of malondialdehyde, a biochemical measure of levels of oxidative stress in the plants. A subset of plant samples will also be sent to the University of New Mexico where stable carbon isotope analysis will be performed.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Bakshi A, Choi WG, Kim SH, Gilroy S. "The vacuolar Ca2+ transporter CATION EXCHANGER 2 modulates hypoxic signaling and response to flooding in Arabidopsis thaliana." New Phytol. 2023 Dec 240;5:1830-1847. https://doi: 10.1111/nph.19274 ; PMID: 37743731 , Dec-2023